Theorem. *In an integral domain* A, *for every polynomial* P *in the variable* x *with coefficients in* A, $P \in \mathsf{A}[x]$, *and every* $r \in \mathsf{A}$, *there exist unique* $\mathcal{Q} \in \mathsf{A}[x]$ *and* $\mathcal{R} \in \mathsf{A}$ *such that*

$$P(x) = (x - r)\,\mathcal{Q}(x) + \mathcal{R},$$

where

$$\mathcal{R} = P(r).$$

○ **1. The existence**

Let $P(x)$ be a polynomial in $\mathsf{A}[x]$ of degree n,

$$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = \sum_{i=0}^{n} a_i x^i,$$

where each coefficient $a_i \in \mathsf{A}$. Evaluating it at $x = r$ gives

$$P(r) = a_n r^n + a_{n-1}r^{n-1} + \cdots + a_1 r + a_0 = \sum_{i=0}^{n} a_i r^i.$$

Consider the difference

$$P(x) - P(r) = \sum_{i=0}^{n} a_i x^i - \sum_{i=0}^{n} a_i r^i.$$

By the linearity of summation, the two expressions can be combined term-by-term. Moreover, the constant terms for $i = 0$ cancel out ($x^0 = r^0 = 1 \Rightarrow a_0 x^0 - a_0 r^0 = a_0 - a_0 = 0$). The difference $P(x) - P(r)$ becomes

$$\begin{aligned}\sum_{i=0}^{n} a_i x^i - \sum_{i=0}^{n} a_i r^i &= (a_0 - a_0) + \sum_{i=1}^{n}\Big(a_i x^i - a_i r^i\Big)\\ &= \sum_{i=1}^{n} a_i\big(x^i - r^i\big).\end{aligned}$$

Each term $x^i - r^i$ in the above sum is a difference of two i-th powers. The following lemma on the factorization of $x^n - c^n$ for any $n \geq 1$ shows that every such difference can be factored by $(x - r)$. Applying the lemma term-by-term exposes a common factor $(x - r)$ across all the terms, so it can be factored out of the entire expression.

Lemma. *For any integer $n \geq 1$, the difference of two n-th powers, $x^n - c^n$, can be factored by $(x-c)$. That is, there exists a polynomial $Q_{n-1}(x,c)$ such that*

$$x^n - c^n = (x-c)\,Q_{n-1}(x,c).$$

○ (by mathematical induction on n)

The **base case** $n = 1$ holds since

$$x^1 - c^1 = (x-c)\,(1),$$

where the constant $1 = Q_0(x,c)$ is a polynomial as well.

Inductive Hypothesis. Assume for some arbitrary positive integer $k \geq 1$ there exists a polynomial $Q_{k-1}(x,c)$ such that

$$x^k - c^k = (x-c)\,Q_{k-1}(x,c).$$

Induction Step. To show that the lemma holds for $k+1$ (assuming it holds for k), consider

$$x^{k+1} - c^{k+1},$$

that can be rewritten by adding $0 = -xc^k + c^k x$ and grouping as

$$x^{k+1} - xc^k + c^k x - c^{k+1} = \left(x^{k+1} - xc^k\right) + \left(c^k x - c^{k+1}\right).$$

Factoring out the common components from each group gives

$$x\left(x^k - c^k\right) + c^k(x-c).$$

Applying the inductive hypothesis to substitute $(x-c)\,Q_{k-1}(x,c)$ for $\left(x^k - c^k\right)$ yields

$$x(x-c)\,Q_{k-1}(x,c) + c^k(x-c).$$

Factoring out the common multiplier $(x-c)$ results in

$$(x-c)\Big(x\,Q_{k-1}(x,c) + c^k\Big),$$

where $x\,Q_{k-1}(x,c) + c^k = Q_k(x,c)$ is a polynomial — since $Q_{k-1}(x,c)$ is a polynomial by the inductive hypothesis and polynomials are closed under multiplication by x and addition of constants.

This completes the induction step, proving the lemma for all $n \geq 1$. ●

Explicitly, the $Q(x, c)$ polynomials are

$$
\begin{aligned}
&1 && = Q_0\\
&x(1) + c^1 = x + c && = Q_1\\
&x(x + c) + c^2 = x^2 + xc + c^2 && = Q_2\\
&x\big(x^2 + xc + c^2\big) + c^3 = x^3 + x^2c + xc^2 + c^3 && = Q_3\\
&x\big(x^3 + x^2c + xc^2 + c^3\big) + c^4 = x^4 + x^3c + x^2c^2 + xc^3 + c^4 && = Q_4\\
&\cdots\\
&x\big(x^{n-2} + x^{n-3}c + \cdots + xc^{n-3} + c^{n-2}\big) + c^{n-1} = x^{n-1} + x^{n-2}c + \cdots + xc^{n-2} + c^{n-1} && = Q_{n-1}\\
&x\big(x^{n-1} + x^{n-2}c + \cdots + xc^{n-2} + c^{n-1}\big) + c^n = x^n + x^{n-1}c + \cdots + xc^{n-1} + c^n && = Q_n
\end{aligned}
$$

In more compact notation with the summation symbol

$$Q_n(x, c) = x^n + x^{n-1}c + \cdots + xc^{n-1} + c^n = \sum_{i=0}^{n} x^{n-i}c^i$$

and

$$xQ_{n-1}(x, c) + c^n = x\sum_{i=0}^{n-1} x^{n-1-i}c^i + c^n = \sum_{i=0}^{n-1} x^{n-i}c^i + c^n = \sum_{i=0}^{n} x^{n-i}c^i = Q_n(x, c)\,.$$

The exact factorization is $\quad x^n - c^n = (x - c)\sum_{i=0}^{n-1} x^{n-1-i}c^i.$

Returning to $$P(x) - P(r) = \sum_{i=1}^{n} a_i\big(x^i - r^i\big),$$

the lemma yields*

$$x^i - r^i = (x - r)\,Q_{i-1}(x, r)$$

for every $i \geq 1$. Substituting these factorizations term-by-term gives

$$P(x) - P(r) = \sum_{i=1}^{n} a_i\big(x - r\big)Q_{i-1}(x, r) = \big(x - r\big)\sum_{i=1}^{n} a_i\,Q_{i-1}(x, r).$$

Since each $Q_{i-1}(x, r)$ is a polynomial in $\mathsf{A}[x]$ and polynomial sums are polynomials, $\sum_{i=1}^{n} a_i\,Q_{i-1}(x, r)$ belongs to the same polynomial ring $\mathsf{A}[x]$. Define

$$\sum_{i=1}^{n} a_i\,Q_{i-1}(x, r) = \mathcal{Q}(x) \in \mathsf{A}[x]\,.$$

Hence $$P(x) - P(r) = \big(x - r\big)\,\mathcal{Q}(x)$$

* From a deeper algebraic viewpoint, $Q_{i-1}(x, c)$ from the lemma belongs to the polynomial ring $\mathsf{A}[x, c]$. Replacing the indeterminate c by the element $r \in \mathsf{A}$ is the evaluation homomorphism $\mathsf{A}[x, c] \to \mathsf{A}[x]$. Thus $Q_{i-1}(x, r)$ is a polynomial in $\mathsf{A}[x]$.

or equivalently $$P(x) = \bigl(x - r\bigr)\mathcal{Q}(x) + P(r).$$

The latter exhibits the required representation with $\mathcal{R} = P(r) \in \mathsf{A}$. Ergo there exist $\mathcal{Q} \in \mathsf{A}[x]$ and $\mathcal{R} \in \mathsf{A}$. ●

○ **2. The uniqueness**

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